CRU

Coordinate Rotation Unit

CORDIC-Based Trigonometric Co-Processor for StarCore-1

OrbitEdge-1 · Attitude Determination

EEE4120F HPES | Milestone 2 | 2026

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18 cycles
Bounded Latency

6× speedup
vs SW CORDIC

0 mult.
No Multipliers

16/16 pass
Gold Std. Tests

THE PROBLEM

Attitude Determination

StarCore-1 computes sin/cos every control-loop iteration for quaternion-based orientation. Trig is the dominant workload.

Software Cost

~108 instruction cycles per sin/cos pair on StarCore-1 — too slow for real-time attitude control at mission frequency.

No FPU, No LUT

Radiation-tolerant FPGAs cannot use a floating-point unit or large LUT. CRU uses only shift-and-add in 18 cycles.

Solution: Dedicated CORDIC hardware — 6× speedup, zero multipliers, bounded 18-cycle latency

SYSTEM ARCHITECTURE

StarCore-1 CPU Core

- 16-bit RISC · Harvard arch
- Single-cycle execution
- Opcode 1010 → CRU

START →
← VALID
← SIN/COS

CRU Co-Processor

- CORDIC FSM (IDLE/ITER/DONE)
- 16 shift-and-add iterations
- 16-entry atan ROM
- Q1.15 fixed-point output

Opcode 1010
Co-Proc Bus
MMIO / Mem

PERFORMANCE COMPARISON

Metric	SW CORDIC	CRU
Cycles / sin+cos	~108	18
Multiplier needed	Yes	No
LUT table size	Large	16 entries
Latency bounded?	No	Yes — 18 cy
Speedup (trig op)	1×	6×
Overall speedup*	1×	~3.2×